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Dear Professor Minlan Yu and the Search Committee,

I am writing to apply for the **Postdoctoral Fellowship in Networking Support for Machine Learning** at the Harvard John A. Paulson School of Engineering and Applied Sciences. The opportunity to work closely with Professor Minlan Yu on the systems infrastructure powering large-scale AI, including distributed inference, collective communication, and emerging agentic workflows, aligns directly with my research trajectory over the past several years, and I am thrilled at the prospect of driving this agenda forward in the group.

I recently completed my Ph.D. in Computer Science at the National University of Singapore, advised by Jialin Li, where I am currently a Postdoctoral Research Fellow. My research develops cross-layer co-designs spanning programmable hardware, systems software, and machine learning. I design protocols and system stacks that offload and optimize functions across layers on which today's large-scale AI training and inference run. My research is grounded in both academic rigor and the realities of industry datacenters at scale. Through two research internships at Microsoft Research and one at AMD, I worked directly with the teams that operate large-scale infrastructure, learning which bottlenecks actually matter in production and which solutions are practically deployable. This grounding shapes both the problems I choose and the systems I build: I aim to solve challenges that arise from real deployment demands, not just testbed benchmarks.

This approach has consistently produced work at top-tier systems and networking conferences. I resolved the single-sequencer bottleneck that had kept network ordering solutions impractical for real-world deployment, by distributing ordering across multiple programmable switches (Hydra, NSDI '23). In cluster-scale transport, I co-developed the first RDMA load balancer for datacenter networks, performing fine-grained rerouting while restoring packet order entirely inside the network fabric (ConWeave, SIGCOMM '23). In a long-term collaboration with Microsoft Research, I designed a new dynamic load balancing architecture for frontend servers, built on inter-server TCP connection migration at microsecond timescales (Capybara, to appear in SIGCOMM '26). Alongside these hardware-software co-design solutions, I currently lead an ML-for-Systems thrust: as static configuration cannot keep pace with the microsecond-scale workload dynamics of modern datacenters, Autokernel (APSys '25) integrates machine learning as a first-class citizen in the dataplane OS for autonomous configuration tuning.

Systems for AI is not a simple extension of classical datacenter networking; it is a distinct agenda that requires re-examining assumptions across networking, systems, and workloads simultaneously. Your group has been one of the most consistent voices articulating that view, and my technical foundation maps onto three assets I would bring to the group:

- **RDMA and collective-communication expertise.** ConWeave (in-network RDMA load balancing) and RecoNIC (RDMA offload on FPGA SmartNIC) give me first-hand familiarity with the RDMA fabric on which today's collective-communication libraries (NCCL, RCCL, MSCCL) run, the exact substrate where distributed training and inference performance is determined.
- **Programmable-switch and SmartNIC systems.** I have designed and implemented systems on programmable switches (Hydra, ConWeave, Capybara) and on FPGA-based SmartNICs

(RecoNIC), covering the in-network coordination, offload, and reordering toolchain that Systems-for-AI increasingly leverages.

- **ML-for-Systems co-design.** My ongoing collaboration with Microsoft Research on Autokernel treats ML as a dataplane tuner with explicit attention to inference cost and safety. Extending this framing more broadly across Systems for AI, including collective-communication scheduling, KV-cache-aware routing for distributed inference, and network primitives for agentic AI workflows, is a direction I am excited to explore.

Beyond research, I take mentoring and teaching seriously. I have closely mentored four research students at NUS: a research assistant on Capybara, two undergraduates on kernel-bypass network stack optimization, and a Master's student whose thesis built on Hydra. Beyond these direct mentoring relationships, I actively help many junior PhD students with their research. Across five semesters I served as a teaching assistant for courses in operating systems, distributed systems, cryptography, and introductory programming. I look forward to taking on the mentorship role that this fellowship carries for undergraduates and graduate students in the Yu group.

I would be delighted to discuss how my background can contribute to Professor Yu's group and to the broader Systems-for-AI research at Harvard SEAS, and I would welcome the opportunity to meet at the committee's convenience.

Thank you for considering my application.

Sincerely,

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